

DIGITAL FORENSIC EXAMINATION REPORT

UNMANNED AERIAL VEHICLE (UAV) FLIGHT DATA & TELEMETRY EXTRACTION

1. INVESTIGATIVE CASE ADMINISTRATION

Case Reference:	CASE-TEST	Evidence Item ID:	tmp_test.bin
Examining Agency:	Cyber Forensic Investigation Laboratory (CFSL / State Police)	Lead Examiner:	Tester
Examination Date:	2026-09-05 15:55:53 UTC	Forensic Engine:	Pushpak Indigenous Drone Forensics Toolkit (v1.0.0-phase1 (Build 2026.09))

2. EVIDENCE INTEGRITY & CHAIN OF CUSTODY

Source Evidence File:	C:\Users\pawan\Desktop\df\tmp_test.bin
File Size:	1,715 bytes
SHA-256 Digest:	330f9a7eea6e16cb694c344913161372174e7624bf9066a855d46632049ef885
BLAKE3 Digest:	5ca5a1e38669da497980d724fcebca142b385459525a1a4646e9fd1609ebc6c9
Chain Verification:	VERIFIED INTACT (100% Cryptographic Match)

3. FLIGHT MISSION ENVELOPE & TELEMETRY PROFILE

Platform / Architecture:	ARDUPILOT	Aircraft Model:	N/A
Hardware Serial No:	N/A	Total Telemetry Records:	37 events
Takeoff Time (UTC):	2025-01-20 03:46:22 UTC	Termination Time (UTC):	2025-01-20 03:46:31 UTC
Total Flight Duration:	9.0 seconds (0.1 mins)	Total GPS Waypoints:	10 points
Max Altitude (AGL):	100.0 meters (328.1 ft)	Peak Ground Velocity:	12.5 m/s (45.0 km/h)

4. FORENSIC ANOMALY FINDINGS & THREAT ASSESSMENT

Anomaly Type	Severity	Timestamp (UTC)	Location / Alt	Technical Finding Narrative
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:23 UTC	19.0765, 72.8782 Alt: 20.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:24 UTC	19.0770, 72.8787 Alt: 30.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:25 UTC	19.0775, 72.8792 Alt: 40.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s

GPS SPOOFING OR TELEPORTATION	HIGH	03:46:26 UTC	19.0780, 72.8797 Alt: 50.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:27 UTC	19.0785, 72.8802 Alt: 60.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:28 UTC	19.0790, 72.8807 Alt: 70.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:29 UTC	19.0795, 72.8812 Alt: 80.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:30 UTC	19.0800, 72.8817 Alt: 90.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:31 UTC	19.0805, 72.8822 Alt: 100.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s

5. FORENSIC CONFIDENCE & UNCERTAINTY ASSESSMENT

Overall Forensic Confidence Rating: HIGH (HIGH COURT ADMISSIBLE) - 99.0%
In accordance with ISO/IEC 27037:2012 guidelines, this assessment explicitly defines the mathematical certainty and evidentiary boundaries of the reconstructed flight profile.

Forensic Dimension	Confidence Level	Evidentiary Basis & Limitations
Bit-Stream Integrity	ABSOLUTE (100%)	Dual SHA-256 + BLAKE3 hashes match original media bit-stream.
Spatial Trajectory	HIGH (99.0%)	Avg Satellites: 14.0 Avg HDOP: 0.80 Multipath risk low.
Temporal Synchronization	HIGH (99.0%)	Correlated with GPS atomic UTC reference clock.
Evidence Limitations	DOCUMENTED	Barometric altitude uncorrected for local QNH pressure.

6. CHRONOLOGICAL MILESTONE TIMELINE

Timestamp (UTC)	Event Type	Mode / State	Forensic Evidence Context
2025-01-20 03:46:21	MODE CHANGE	AUTO	MODE
2025-01-20 03:46:21	ARM DISARM	AUTO	MSG
2025-01-20 03:46:22	GPS FIX	AUTO	GPS
2025-01-20 03:46:31	GPS FIX	AUTO	GPS
2025-01-20 03:46:32	RTH TRIGGER	AUTO	MSG
2025-01-20 03:46:32	MODE CHANGE	RTL	MODE
2025-01-20 03:46:36	ARM DISARM	RTL	MSG

7. CERTIFICATE OF AUTHENTICITY (LEGAL ADMISSIBILITY)

I hereby certify under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (corresponding to Section 65B of the Indian Evidence Act, 1872) and ISO/IEC 27037 standards that the electronic records, telemetry chronologies, and mathematical calculations contained within this report were generated by lawful digital forensic acquisition. The computing systems and software tools used to parse this evidence operated properly throughout the examination with zero alteration to original electronic memory. The cryptographic digests (SHA-256 and BLAKE3) confirm absolute bit-stream integrity of the exhibited evidence.

Examiner Signature: _____

Date: 2026-09-05

Name: Tester

Title: Lead Digital Forensics Examiner

Agency: Cyber Forensic Investigation Laboratory (CFSL / State Police)

Official Seal: [AFFIX LABORATORY SEAL]